Technical report: RöstiBridge

- **Track:** Track 2A, UZH: Detecting Cross-Lingual Semantic Differences in Swiss Government Websites
- **Event:** Online
- **Team:** RedPandas: Christopher Yazigi, Giovanni Yazbeck, Alfredo Pio Yazbeck
- **Demo:** Firefox extension in `src/extension` (`dist/roestibridge-0.1.5.xpi`); screenshots in `docs/`

1. Summary

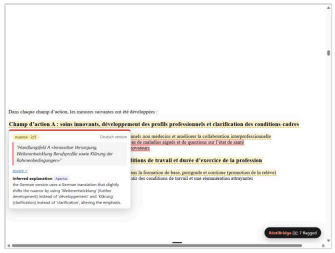
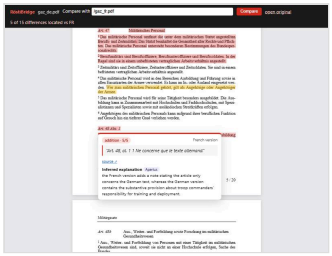
Swiss authorities publish the same page in up to four languages, and the versions do not always say the same thing: sentences are dropped, numbers or deadlines change, or "must" becomes "may". RöstiBridge, a bridge over the *Röstigraben*, brings token-level recognition of semantic differences (RSD) to the reader. It is a Firefox extension that highlights passages of an official page (federal, cantonal, RTS news, or a parliamentary PDF) whose meaning may differ from the reference language version. Hovering a highlight shows the other version's excerpt as a citation and an Apertus explanation.

The same single Apertus prompt is benchmarked on SwissGov-RSD [1]. Apertus-v1.5-70B reports difference spans, which are projected onto tokens; 3-sample self-consistency averages them. On the full dev split (504 document pairs) the official UZH scorer gives a macro token-level Spearman of **0.193** (DE 0.210, FR 0.165, IT 0.204), against 0.095 for a model-free number-mismatch baseline. On the 120-document subset used for development the gain is from 0.096 for raw single-sample spans to 0.131. The submission files in `predictions/submission/` cover all 224 documents per language in the official full-set order.

2. Architecture

```
page / PDF in language X ──► extract blocks ──► find counterpart in language Y
  (content script / pdf.js)                      hreflang · switcher · /de/⇔/fr/ URL
                                                 · Apertus path matching (admin.ch CMS)
                                                 · OpenParlData affair siblings (parlament.ch PDFs)
                                                 · Bing-News RSS + Apertus story matching (RTS↔SRF)
                                                                  │
            └──────► background.js: chunk aligned blocks (≈3.5k chars, proportional
                     window ±4 blocks) ──► Apertus (shared/prompt.json) ──► JSON
                     {a, b, type, severity, explanation} ──► cache (storage.local)
          ◄──────── highlight `a` excerpts (whitespace-agnostic matching across nodes /
                     text-layer spans) + hover card (citation `b` + explanation)
```

- **Federal vs. cantonal logic** (`shared/sites.js`):
  - Federal domains are compared with FR and DE (whichever the page is not in).
  - Bilingual cantons (BE/FR/VS/GR) are compared with the other official language.
  - Monolingual cantons are compared with their official language.
- **Only real differences are shown:** a reported difference whose two excerpts are the same string, or whose "missing" text does appear in the other version (place names, datelines, numbers, acronyms), is dropped before display. The cache is versioned, so a change to this filtering never serves stale results.
- **Explanations in the reader's language** (EN / DE / FR / IT, set in the options): the request states the explanation language explicitly, and "TEXT A / TEXT B" in Apertus' answer become "the French version", "die deutsche Fassung", etc. For English the request stays byte-identical to the frozen benchmark prompt.
- **Robustness:** every Apertus request has a 120 s timeout and 3 retries. Each section of a page is checked on its own, so one slow answer never discards the others. Only complete results are cached, and errors show their reason (e.g. `invalid API key`). The options page tests the key when you save it.
- **Key handling:** the API key lives in `browser.storage.local` and is only read by the background script. Page scripts never receive it.
- **Benchmark pipeline** (`src/roestibridge`): a stdlib-only Python client (`llm.py`), span → token projection (`rsd.py`), the metric (`metrics.py`, a faithful re-implementation of the official `evaluate_predictions_admin.py` pooling) and a CLI. It reads the **same** `prompt.json` as the extension, and a unit test enforces this.

Federal page: hover card	Parliamentary PDF: hover card
	

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` (CSCS-hosted, OpenAI-compatible endpoint). `Apertus-v1.5-8B` can be selected in the options for on-prem use.
- **How it is used:** prompting-only inference with JSON output (`response_format: json_object`, with a fallback when the endpoint does not support it). Apertus is used for:
  1. difference detection with severity and explanation (core RSD);
  2. matching translated URL slugs on the new admin.ch CMS;
  3. turning an RTS headline into German search queries and deciding whether an SRF article is the same story;
  4. picking the translated sibling of a parliamentary document.
- **Prompt:** `src/extension/shared/prompt.json` (`rb-prompt-v1`). The system prompt defines what counts as a difference (omission/addition, numbers/dates, obligation/scope, legal terms, actionable nuance) and what does not (grammar, word order, free translation). It asks for exact, short excerpts.
- **Benchmark settings:** temperature 0 for the first sample and 0.7 for samples 2–3, `max_tokens` 2000, one call per document pair. Token usage (from the API) is about 0.82M prompt and 0.25M completion tokens for the 3-sample test run, about 4.9k prompt + 1.5k completion tokens per document pair (`raw_*.jsonl` → `usage`).
- **No other model is used** anywhere in the pipeline or for evaluation.

4. Data

- **SwissGov-RSD** [1] (`ZurichNLP/SwissGov-RSD` on Hugging Face) is downloaded at run time into `data/swissgov-rsd/` and is not committed.
- Development used a deterministic, evenly spaced **40-document subset of the dev split per language** (`--limit 40`, 120 pairs) to stay within the API budget. After freezing, the remaining dev documents were predicted with the same setting, because the UZH Q&A asks for predictions in the full-set order (224 per language).
- **The test split was never evaluated or inspected:** `evaluate --split test` exits with an error.
- Test predictions were produced once with the prompt and post-processing frozen on dev.
- The extension processes public government pages on the fly and stores nothing beyond a local cache of model answers.
- Note: the dataset's DE/FR/IT texts contain U+FFFD replacement characters in place of accented letters (e.g. `L`  `approvisionnement`). The prompt tells the model to copy excerpts verbatim "even if characters look garbled", and the projection treats as a word character.

5. Evaluation

Metric: the official one. For each language pair, all tokens of side A and side B with gold $\neq -1$ are pooled (predictions are padded or truncated to the gold length), global Spearman is computed, and the macro average is taken over EN–DE/FR/IT. Our implementation matches `scipy.stats.spearmanr` on the pooled vectors. It also matches the official `scripts/evaluate_predictions_admin.py` exactly on the full dev split (DE 0.210 / FR 0.165 / IT 0.204; Kendall's τ 0.197 / 0.154 / 0.187).

Setup (dev subset, 40 docs/pair)	DE	FR	IT	macro
Number-mismatch baseline (no LLM)	0.102	0.068	0.080	0.083
Apertus-70B, 1 sample, all spans at severity/5	0.136	0.056	0.096	0.096
+ drop severity 1, omission/addition = 1.0	0.157	0.069	0.100	0.109
+ number-mismatch prior (max)	0.159	0.074	0.104	0.112

Setup (dev subset, 40 docs/pair)	DE	FR	IT	macro
+ 3 samples, averaged (submitted)	**0.191**	**0.077**	**0.125**	**0.131**
Full dev split (168 docs/pair), number baseline	0.132	0.065	0.087	0.095
Full dev split (168 docs/pair), submitted setting, official scorer	**0.210**	**0.165**	**0.204**	**0.193**

Ablations (no extra API calls; `python -m roestibridge.ablate`):

Variant	Macro	Result
Binary span scores instead of severity	0.095	Worse
Damping long spans by length	±0.001	No gain
Paragraph-context smoothing	0.099	No gain

Error analysis (FR, 1 sample): - **Precision is the lever.** Of the tokens we mark, 226 are true positives and 1,752 are false positives. Apertus over-reports "different wording, same meaning" spans at low severity, which motivated the severity ≥ 2 cut. Many missed gold differences (1,119 tokens) are small structural omissions such as link labels and list items. - **Span projection is reliable:** 376 of 382 returned excerpts were located in the token sequence. - **FR is the hardest pair.** The FR documents are longer (461 vs 363 tokens on average for DE) and the FR texts carry the most `&#xFFFD;` corruption.

Extension checks. These ran end to end in Firefox 158 (headless, WebDriver BiDi via puppeteer-core) on the packaged extension against live sites on 16 Oct 2026. Apertus-70B was used throughout. The counts vary between runs because the model is not deterministic.

Case	Result
bfe.admin.ch/fr → DE	3–4 highlighted, e.g. a paragraph on the IAEA General Conference that is missing in the German version
be.ch/fr → DE (bilingual canton)	1 highlighted, a tone shift ("beau palmarès" vs "brillieren")
bag.admin.ch FR → DE	Counterpart found via Apertus slug matching (`agenda-soins-de-base` → `agenda-grundversorgung`); up to 9 differences, mostly list items
ch.ch/it → DE and FR (federal: both)	Both versions compared; the homepages agree (0 differences)
RTS Amaudruz article → SRF	SRF story found and linked ("SVP-Nationalrätin Céline Amaudruz will Bundesrätin werden"); 8–9 differences highlighted
parlament.ch PDF link (DE)	Opened in the RöstiBridge viewer; FR counterpart found via OpenParlData affair 234959 (`Kinderabzug = 12'000 Franken` → `déduction pour enfants = 12 000 francs`); the tables agree (0 differences)
Fedlex consultation PDF (DE↔FR)	16 differences, including articles where the French reads "Ne concerne que le texte allemand" while the German changes the provision; 10 of 16 placed on the text layer

6. Limitations

- **Document-level RSD is hard for prompted LLMs,** consistent with [1]. The dev subset is small (120 pairs), so the post-processing choices may be slightly overfit.
- **Extension chunk alignment is proportional (± 4 blocks).** When a page has very different structures across languages, content outside the window can be reported as "missing". Apertus sometimes quotes the wrong side or paraphrases. The locator then tries the other excerpt and the first or last six words of the quote.
- **Counterpart discovery depends on site conventions.**
  - The admin.ch slug matcher needs both homepages to list the page.
  - RTS→SRF depends on a third-party news search (Bing News RSS, DuckDuckGo fallback), and SRF is independent journalism, not a translation. Differences there mean coverage differences, not translation errors.
- **Language metadata:** OpenParlData labels some French parlament.ch documents as `it`. We therefore ask Apertus to pick by title rather than trusting the language field.
- **Packaging:** the extension is unsigned (temporary add-on) and uses Manifest V2 for blocking `webRequest` (PDF interception).

7. Reproducibility

- `make run` (Docker, no GPU) runs four steps:
  1. unit tests;
  2. re-scoring of the shipped raw Apertus outputs for dev (exactly the numbers above, no API needed);
  3. the model-free baseline;
  4. rebuilding the submitted test predictions from the shipped raw outputs.

  With `LLM_API_KEY` set it also runs a live Apertus prediction and evaluation on `LIVE_LIMIT` dev documents per language.
- `make predict-test` regenerates the test predictions (168 × 3 calls).
- Apertus output at temperature 0 is not perfectly deterministic on the hosted endpoint. That is why the raw responses are shipped and all scores are recomputed from them.

8. Next steps

- **Sentence alignment before prompting.** Align with the LLM or embeddings, then ask per aligned pair. This should improve both recall and the "missing" false positives.
- **Calibrate severities per language** on dev.
- **Try 8B with more samples** for an on-prem budget.
- **Ship a signed MV3 build:** declarativeNetRequest for PDFs, plus a "report this discrepancy to the authority" button that drafts a correction request with the citation.

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References

[1] Wastl et al. *SwissGov-RSD: A Human-annotated, Cross-lingual Benchmark for Token-level Recognition of Semantic Differences Between Related Documents.* arXiv:2512.07538 / ACL 2026. https://arxiv.org/abs/2512.07538